

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. II Semester Examination, 2026

Regulation: R20 | University: JNTUK

Operating Systems (21CS41)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Operating System Overview & Structures]

- a) Explain the functions of an operating system and draw the layered structure.
- b) Compare long term, medium term and short term schedulers.

UNIT - 2 [Process Management & Scheduling]

- a) Solve the given set of processes using FCFS, SJF and Round Robin and compare average waiting time.
- b) Explain the inter process communication mechanisms.

UNIT - 3 [Concurrency, Synchronization & Deadlocks]

- a) Explain the producer consumer problem using semaphores.
- b) What is a race condition? Explain with an example.

UNIT - 4 [Memory Management & Virtual Memory]

- a) Compare paging and segmentation.
- b) Explain LRU and FIFO page replacement algorithms with an example.

UNIT - 5 [File Systems, I/O & Protection]

- a) Compare contiguous, linked and indexed file allocation methods.
- b) Explain disk scheduling algorithms FCFS, SCAN and C-SCAN with an example.

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